

Critical characteristics of innovation

Steps we took from collecting the original lists

1. **Formulation of Question:** We sent the question to 20 people and received 17 answers.

"Föreställ dig en framtida hemmarobot som skulle kunna hjälpa till med att städa, fungera som en alexa, beställa mat, hålla koll på din hälsa med hjälp av data om bl.a. dina inköp, vikt, puls, som kopplar till din smartwatch. Vad skulle få dig att börja använda en sådan produkt i din vardag? Vilka fördelar eller risker finns det? Vilka tankar dyker upp när du föreställer dig produkten?"

2. **Collection of answers:** The answers were collected into one document in raw text form
3. **Categorization and counting the frequency:** Went through the answers and categorized them into different styles of beliefs according to the answers of the participants and then grouped them into belief groups and counted the frequency.
4. **Identification of modal beliefs:** Analysed the responses and identified the most common beliefs.
5. **Creation of final list:** Compiled a list of 7 modal salient beliefs which we would create a set of questions for

Group	Home robot	Frequency	Belief number
6	Price	8	1
2	I do not need it (now and in the future)	9	2
5	Health (monitoring etc.)	8	3
7	Scary/alien, feels like a strange idea	7	4
1	Data security (don't want anyone to have my health info)	10	5
6	Upgradable	1	6
	Incompatible to ones home	7	7
	companion	2	9
2	I do not need it now(maybe in the future (as older))	10	10
3	Saving time	10	11
	Sympatisk	1	12
7	Removes free will	5	13
7	Monitoring feel stressful (negative)	4	8
4	performance/features	13	14
	reliability	2	15

Modal salient beliefs:

1. Data security (don't want anyone to have my health info)
2. I do not need it
3. Saving time (convenience)
4. performance/features
5. Health (monitoring etc.)
6. Price
7. Aversions to new technology

Questions for assessing the relative impact of your identified modal salient beliefs

1. Data security (Scale Totally Disagree/ Totally Agree -3,3)

- I would be comfortable with a company using data about my health to develop a service I'm using
- I'm not worried about data about myself, gathered by services i use, being compromised or leaked in a data breach
- I am comfortable with a company providing me more accurate advertisements, based on data gathered about myself
- I would be comfortable with my data being collected, if I knew it was stored within the EU.

2. I don't need the product or service (Scale Totally Disagree/ Totally Agree -3,3)

- I would consider testing a device I don't need right now, for free, if someone i knew found it useful
- I would be open to stop using certain home appliances, if they were replaced by a robot

3. Saving time (Scale Totally Disagree/ Totally Agree -3,3)

- This would help me save a lot of time
- I could not adjust this to my daily life
- I consider this as useful

4. Performance/Features (Scale Totally Disagree/ Totally Agree -3,3)

- I would like to be able to add and delete features in the robot myself.
- I would prefer setting health goals myself
- I would prefer to buy the product for myself instead of leasing it
- The robot needs to be possible to turn off
- This would elevate my quality of life

5. Health (Scale Totally Disagree/ Totally Agree -3,3)

- I think having a robot that reminds me about my health routines (medication, hydration, exercise) would help me.
- Health monitoring is one of the most valuable features that the robot would offer me.
- The ability of the robot to help me make better health choices would make me feel more more in control of my wellbeing

- A robot that monitors my health would motivate me to take better care of myself.
- If the robot's health features were customizable to my needs, I would see it as a valuable feature.

6. Price (Scale Totally Disagree/ Totally Agree -3,3)

- A high cost would prevent me from considering buying this robot
- I would be more likely to buy the robot if I knew that I could upgrade the robot's hardware over time.
- I would prefer a monthly subscription over a one-time purchase for the robot
- I would find it easier to justify the cost of the robot if it replaced several other devices i am currently using

7. Aversions to new technology (Scale Totally Disagree/ Totally Agree -3,3)

- I would be fine with a device using constantly recording cameras in my home
- I would appreciate constant notifications about my health
- I would be fine with relinquishing responsibility for my health to a robot
- I have no problem helping delivery robots if they are stuck, e.g. on the sidewalk
- I would be okay with redecorating or making adjustments to my home in order to accommodate a robot

Questions for measuring the behavioral criterion

2. Need for the service

- Compared to my current methods of cleaning, consuming media and grocery shopping, using a robot that automates these things within the next 6 months would be
 - Invaluable/Redundant
 - Liberating/Limiting
 - Relaxing/Stressful
- I am currently using or considering ordering:
 - A smart home device (e.g. Alexa)
 - A smart watch
 - A robot vacuum

4. Performance/Features

- I would find this useful in following features
 - Outside home digital systems
 - Inside home digital systems
 - Inside home physical duties

5. Health

- Compared to my current form of monitoring my health, using a robot that support my health tracking at home in the next 6 months would be
 - Pleasant/Unpleasant
 - More convenient/Less convenient
 - Comfortable/Uncomfortable

6. Price

- Compared to my current budget for home devices, buying the robot at an affordable price within the next 6 months would be
 - Worth it/ Not worth it
 - Financially reasonable/Financially risky
 - Affordable within my budget/too expensive for my budget